

Exercise Sheet 3

Advanced Microeconomics: Economic Decision Making

Question 1

Let u be three times continuously differentiable with $u' > 0$, and let ε be a zero-mean risk with variance σ^2 and small support around 0; initial wealth x . Define the risk premium ρ by $u(x - \rho) = \mathbb{E}[u(x + \varepsilon)]$.

- (a) Expand both sides to second order in small quantities and derive $\rho \approx \frac{1}{2}r_A(x)\sigma^2$ where $r_A = -u''/u'$. [15]
- (b) For proportional risks $\varepsilon = x\tilde{\varepsilon}$ with $\tilde{\varepsilon}$ zero-mean of variance $\tilde{\sigma}^2$, deduce $\rho/x \approx \frac{1}{2}r_R(x)\tilde{\sigma}^2$. [5]
- (c) In one sentence each: why is the premium quadratic in the size of the risk, and what did this quadratic imply for the course's ex-ante cost-of-inflation numbers? [5]

Question 2

A worker's real income lottery is: uninsured, -1% w.p. 0.75 and -9% w.p. 0.25; with an indexation contract, $+0.5\%$ w.p. 0.75 and -1.5% w.p. 0.25. Separately, the contract raises the probability of a job loss worth -40% of income by 1pp in the bad state; treat this as an expected cost computed *outside* the lottery. CRRA, $r = 2$.

- (a) Compute both certainty equivalents and the gross insurance value ΔCE . [12]
- (b) Decompose ΔCE into a mean part and a risk part, and interpret the split. [8]
- (c) Net off the job-cost term and state the final valuation with its three components. [5]

Question 3

$u(x) = -e^{-ax}$ with $a = 0.01$; initial wealth w ; the agent faces a fair coin paying ± 100 .

- (a) Derive the exact certainty-equivalent formula $c = w - \frac{1}{a} \ln(\frac{1}{2}e^{a \cdot 100} + \frac{1}{2}e^{-a \cdot 100})$ and evaluate the risk premium. [12]
- (b) Compare with $\frac{1}{2}a\sigma^2$ and explain the direction of the discrepancy. [7]
- (c) Verify explicitly that the premium does not depend on w , and state which behavioural implication of CARA this illustrates (and why it is empirically awkward). [6]